EHMS SYSTEM RECTIFICATION REPORT

Complete Documentation of System Flaws Rectified & Workflow Enhancements

Project Repository: Enterprise-Hospital-Management-System (EHMS)
Status: All Rectifications Applied & Verified
System Scope: Emergency, Inpatient (Bed Management), Triage & Central Billing Engine
Deployment: Vite Dev Server (Port 8443 / Live)
Date of Rectification: November 6, 2026 (Production Dev)

1. Executive Summary

This technical engineering report outlines all system defects reported by the user, root causes identified, and exact programmatic rectifications deployed across the Enterprise Hospital Management System (EHMS). All issues across the Emergency Department, Inpatient Bed Management, Patient Triage, Discharge Clearance, and the Central POS Billing Engine have been systematically resolved, maintaining strict data integrity and real-time counter synchronization.

#	Reported Flaw	Module / Component	Core Cause	Status
1	Duplicate Patient Creation in Emergency	Emergency (ErPage / erDb)	Submitting ER registration triggered non-idempotent inserts, creating twin visits.	RESOLVED
2	Emergency "Send to Billing" Bug & Redirection	Emergency -> Billing Desk	Button was incrementing bill sum artificially & redirecting away from clinical ER desk.	RESOLVED
3	ER "Request Bed" Not Appearing in Inpatient	Emergency -> Bed Management	Handoff alerts were not surfacing on the Inpatient Bed Board for ward bed allocation.	RESOLVED
4	Bed Discharge Showing Incorrect 3 Days	Inpatient Bed Discharge Modal	Checklist API hardcoded length of stay to 3 days regardless of admission timestamp.	RESOLVED
5	Confirm Discharge Not Sending Bill to IP Billing	Bed Release -> IP Billing Counter	Falsy guard checks on bed occupant name dropped claim creation on bed release.	RESOLVED
6	Central Billing Queue Empty / Filter Drops	billingDb / billingSync / Billing.tsx	Aggressive seed purge regex dropped CLM-89xx series; missing INPATIENT dept mapping.	RESOLVED

2. Technical Details & Root Cause Analysis

2.1 Emergency Duplicate Patient Registration

Issue: Registering an emergency patient created two duplicate records simultaneously in the active triage queue.

Root Cause: Registration handler executed parallel dispatch calls to both local state array unshift and erDb database insert without deduplicating by visit number or MRN.

Rectification: Implemented deduplication and idempotent upsert in ErDatabase.createVisit and ErPage registration handler. Active visits check for existing patient ID/name match before creating new triage entries.

2.2 Emergency "Send to Billing" Bug & Redirection Removal

Issue: Clicking "Send to Billing" artificially doubled or accumulated charges on subsequent clicks and unexpectedly navigated staff away from the Emergency desk.

Root Cause: The send-to-billing routine re-appended existing items into line charges and called setModule("billing"), kicking doctors/nurses out of emergency triage.

Rectification: Refactored billing calculation to derive a clean single invoice packet with subtotal matching actual clinical items. Replaced screen redirect with in-page success toast notice while keeping user on ER desk.

2.3 Emergency "Request Bed" to Inpatient Handoff

Issue: Clicking "Request Bed" inside Emergency did not surface the request on the Inpatient Bed Management module.

Root Cause: Bed request notifications were decoupled from the Inpatient Bed Board polling loop; BedManagementPage only queried static hospital wards without listening to transfer alerts.

Rectification: Connected ErDatabase.dispatchBedTransferAlert and er_bed_requests with Inpatient BedDatabase. BedManagementPage now renders pending bed allocation banners, allows 1-click bed assignment from ER, and synchronizes status back to ER.

2.4 Bed Discharge Modal Length of Stay (LOS) Dynamic Calculation

Issue: Discharge checklist modal displayed 3 days stay for patients allocated to a bed on the very same day.

Root Cause: Mock endpoint `/api/beds/:id/discharge-checklist` hardcoded `calculatedDays = 3` and `days: 3` in its room charge breakdown.

Rectification: Replaced hardcoded 3 days with dynamic time difference: `Math.floor((now - admission_date) / 86400000)` with a minimum of 1 day. Made the `Days` table column editable in BedManagementPage so billing staff can manually adjust LOS before finalizing.

2.5 Bed Discharge Bill Dispatch to IP Billing Counter

Issue: Clicking "Confirm Discharge" on an occupied bed discharged the bed but failed to show the room charges bill in IP Billing.

Root Cause: 1) Guard check `selectedBed.patient_name` failed when patient names were stored in `patient_last_name` or `bedOccupantName`. 2) If room charge segments had not loaded, `roomChargeTotal` was 0, silently skipping claim creation. 3) `MODULE_DEPT` in `billingSync.ts` lacked `INPATIENT` mapping.

Rectification: Added resilient fallbacks: `patientName` via `bedOccupantName`, fallback `patientId`, and fallback total (`daily_rate * days`). Created claim with `department: "Inpatient"` (or ICU), created department charge, mapped `INPATIENT: "Inpatient"` in `billingSync.ts`, and dispatched `BILLING_UPDATE_EVENT`.

2.6 Central Billing Engine & Seed Purge Stabilization

Issue: Generated bills were disappearing immediately after creation, leaving queue count at (0).

Root Cause: `BillingDatabase.load()` contained an over-aggressive legacy seed filter regex (`/^CLM-89\d{2}$/`) that matched newly generated claims (`CLM-8938`, `CLM-8939`) and purged them from localStorage.

Rectification: Removed the legacy seed purging regex from `billingDb.ts` so all user-created invoices and claims persist permanently.

3. Modified Codebase Files & Impact Analysis

File Path	Changes	Description & Impact
ehms/frontend/src/pages/BedManagementPage.tsx	+139 / -13	Dynamic LOS calculation from admission timestamp; editable days input; robust occupant name fallback in handleRelease; claim & department charge creation on discharge.
ehms/frontend/src/lib/api.ts	+122 / -9	Checklist endpoint computes LOS from bed admission/allocation date; release bed API (/api/beds/:id/release) generates backup Inpatient Claim & preselection.
ehms/frontend/src/pages/ErPage.tsx	+146 / -18	Prevented duplicate ER registration; fixed Send to Billing calculation without inflation; removed unwanted billing redirect; connected Request Bed to Inpatient Bed Board.
ehms/frontend/src/services/billingDb.ts	+40 / -40	Removed aggressive /^CLM-89\d{2}$/ seed purge from load(); ensured live updates dispatch on claim creation and update.
ehms/frontend/src/services/billingSync.ts	+6 / -0	Added INPATIENT, BILLING_IP, IP_BILLING, BILLING_ER to MODULE_DEPT dictionary to guarantee correct counter routing.
ehms/frontend/src/services/erDb.ts	+38 / -2	Added deduplication for patient visits and alerts; ensured ER bed requests dispatch proper timeline events.
ehms/frontend/src/components/Inpatient.tsx	+3 / -0	Updated Inpatient module routing to maintain synchronization with bed board and admissions.

4. Quality Assurance & Verification Procedures

Test 1: Single Patient Registration in Emergency

Navigate to Emergency module -> Register a patient. Verify that exactly ONE patient card appears in the Emergency Triage Board without duplicates.

Test 2: Emergency "Send to Billing" Verification

Add triage / ER items and click "Send to Billing". Verify toast confirmation appears ("ER Bill Generated: Invoice INV-... sent to Emergency Billing"). Confirm you stay on Emergency screen. Navigate to Billing > Counters > Emergency Billing and verify the bill is waiting in DUES queue.

Test 3: ER Bed Request to Inpatient Handoff

In ER Patient Detail, click "Request Bed". Navigate to Inpatient > Bed Management. Verify that the pending ER transfer alert and bed allocation request appear at the top. Click "Allocate Bed" to assign a bed.

Test 4: Length of Stay (LOS) Verification

Click on the newly allocated bed on the Bed Board. Click "Discharge". Verify the checklist shows "1" Day for today's admission instead of hardcoded 3 days. Verify that the days input field is editable.

Test 5: Confirm Discharge & Inpatient Billing Flow

Click "Confirm Discharge". Verify bed returns to Available status. Navigate to Billing > Counters > IP Billing. Verify the discharged patient appears in the "IP BILLING QUEUE" under DUES, with accurate room charges and OUTSTANDING DUES metric updated.